EXISTENCE, NORMALITY AND UNIQUENESS OF THE SPLITTING FIELD OF A POLYNOMIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The uniqueness isomorphism sigma^{-1} composed with the inclusion realizes 'unique up to (nonunique) isomorphism' of the statement.

Editorial note. The label lemma-splitting-field names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-algebraically-closed (factorization over an algebraic closure), lemma-map-into-algebraic-closure (extension of embeddings), lemma-characterize-normal (normality via equal images of embeddings)

Source reference. Original reference 09GR: sources/stacks-fields/pinned-original-source.tex, source lines 1128–1128. Prerequisite pointer invoked at 2276; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09HQ: sources/stacks-fields/pinned-original-source.tex, source lines 2096–2096. Prerequisite pointer invoked at 2291; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GU: sources/stacks-fields/pinned-original-source.tex, source lines 1240–1240. Prerequisite pointer invoked at 2287; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let F be a field. Let $P \in F[x]$ be a nonconstant polynomial. There exists a smallest field extension E/F such that P splits completely over E. Moreover, the field extension E/F is normal and unique up to (nonunique) isomorphism.*

Proof. Choose an algebraic closure $\overline{F}$. Then we can write $P = c(x-\beta_1)\ldots(x-\beta_n)$ in $\overline{F}[x]$, see Lemma 09GR. Note that $c \in F^*$. Set $E = F(\beta_1, \ldots, \beta_n)$. Then it is clear that E is minimal with the requirement that P splits completely over E.

Next, let E' be another minimal field extension of F such that P splits completely over E'. Write $P = c(x-\alpha_1)\ldots(x-\alpha_n)$ with $c \in F$ and $\alpha_i \in E'$. Again it follows from minimality that $E' = F(\alpha_1, \ldots, \alpha_n)$. Moreover, if we pick any $\sigma : E' \to \overline{F}$ (Lemma 09GU) then we immediately see that $\sigma(\alpha_i) = \beta_{\tau(i)}$ for some permutation

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$\tau : \{1, \ldots, n\} \to \{1, \ldots, n\}$. Thus $\sigma(E') = E$. This implies that E' is a normal extension of F by Lemma 09HQ and that $E \cong E'$ as extensions of F thereby finishing the proof. $\square$